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Assignment for Computer technology

Answers

1. Cpp(sum,product,difference,andquotient)

```
#include <iostream>
using namespace std;
int main() {
    int num1,num2;
    cout<< "Enter two integer values:";
    cin>> num1 >> num2;
    int sum = num1 + num2;
    int product = num1 × num2;
    int difference = num1 - num2;
    // use double for quotient to handle decimal values
    double quotient = ( num2 != 0)?
    ( double) num1 / num2 : 0.0;
    cout << " sum: " << sum << endl;
    cout << " product: " << product << endl;
    cout << " difference: " << difference << endl;
    if (num2 != 0) {
        cout << "quotient: " << quotient << endl;
    } else {
        cout << "Quotient: division by zero undefined" << endl;
    }
    return 0;
}
```

2. Cpp(Area of a circle and square root)

```
#include <iostream>

#include <cmath> // Required M_PI
and sqrt()

using namespace std;

int main() {

    double radius;

    cout << " Enter the radius of the
    circle: ";

    cin >> radius;
```

```

// Use M_PI for pi, often found
in <cmath> or defined manually: const
double pi = 3.14159;
const double pi =
3.14159265358979323846;
double area = pi *pow(radius,2);
double squareRoot = sqrt(area);

cout << "Area of the circle: " <<
area << endl;
cout << " square root of the are: "
<< squareRoot << endl;
return 0;

```

3. Cpp(check for equality of three integers)

```

#include <iostream>
using namespace std;

int main() {
int a, b, c;
cout << " Enter three integer values: ";
cin >> a >> b >> c;

// use the equality operator (==)
if (a == b && b == c) {
cout << "Equal" << endl;
} else {
cout << "Not Equal" << endl;
}

return 0;
}

```

4. Cpp(compare strings)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    String str1,str2;
    Cout << " Enter the first string: ";
    cin >> str2;

    // use the equality operator (==)
    for string comparison
    if (str1 == str2) {
        cout << "Hello!" << endl;
    } else {
        cout << "goodbye! << endl;
    }
    return 0;
}
```

5. Cpp(compare two integers)

```
#include <iostream>
using namespace std;

int main() {
    int val1, val2;
    cout << "Enter two integers values: ";

    cin >> val1 >> val2;

    // use the greater than operator (>)
    if (val1 > Val2) {
```

```
    cout << "first value is greater" << endl;
} else if (val2 > val1) {
    cout << "second value is greater" << endl;
} else {
    cout << " values are equal" << endl;
}
return 0;
}
```

6. Cpp(Uppercase character)

```
#include <iostream>
#include <cctype> // Required for isupper()
using namespace std;

int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;

    // check if the character is an uppercase letter

    if ( isupper(ch)) {
        cout << "Uppercase letter" << endl;
    } else {
        cout << "Not an uppercase letter" << endl;
    }
    return 0;
}
```

7.Cpp(string: :find())

```
#include <iostream>

#include <string>

using namespace std;

int main() {
    String str;
    cout << "Enter a string: ";

    // use getline for strings that may contain spaces
    // cin. ignore(); // If needed after a precious cin >>
    // getline(cin, str);

    cin >> str; // Assuming single word input as typical in assignments

    // string: :find() returns
    String: :npos if the character is not found
    if (str.find('a') != string: :npos)
    {
        cout << "Contains 'a' " << endl;
    } else {
        cout << "Does not contain 'a' " << endl;
    }
    return 0;
}
```

8.Cpp(Odd & Even integer)

```
#include <iostream>
using namespace std;

int main() {
    int number;
    cout << "Enter an integer: ";
    cin >> number;

    // The modulus operator (%) gives
    the remainder of a division
    if (number % 2 == 0) {
        cout << "Even number" << endl;
    } else {
        cout << "Odd number" << endl;
    }
}
```

```

    }
    return 0;
}

```

9. Difference between Assignment and Equality Operators

Operator

Assignment

Equality

Name

Simple Assignment

Equal to

Symbols

=

==

Purpose

Assigns the value of the right operand to the variable on the left operand

Compares the values of two operands.

Examples

Int x = 5;

if (x == 5)

10. Cpp(Ternary operator to Determine Larger of two integers)

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
    int num1 , num2;
    cout << "Enter two integers: ";
    cin >> num1 , num2;
```

```
// The ternary operator determines the larger number
```

```
int larger = (num1 > num2) ? num1 : num2;
```

```
cout << "The larger number is: " << larger << endl;
```

```
return 0;
```

```
}
```

11. Increment and Decrement operators

The increment operator (++) increases a variable's value by 1.

The decrement operator (--) decreases it by 1, both have prefix and postfix forms.

12. Cpp(Logical And)

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
    int number;
    cout << "Enter an integer: ";
    // Condition 1: check if odd
```

```

(remainder 1 when divided by 2)
bool isOdd = (number % 2 != 0);
//Condition 2 : check if divisible by 3
(remainder 0 when divided by 3)
== 0);
//Combine Conditions using Logical
AND (&&)
if (isOdd && isDivisibleBy3) {
    cout << number << " is NOT both
Odd and divisible by 3." << endl;
}
return 0;
}

```

13.Cpp(Bitwise XOR Operator)

```

#include <iostream>
using namespace std;

int main() {
    int x = 10, y = 20;
    cout << "Before swap: x = " << x << ", y = " << y << endl;
    // XOR swap Algorithm
    // Step 1: x = x ^ y
    x = x ^ y; // x is now ( original_x ^ original_y)
    // step 2: y = x ^ y
    y = x ^ y; // y is now ( original_x ^ original_y) ^ original_y = original_x
    // step 3: x = x ^ y
    x = x ^ y; // x is now ( original_x ^ original_y)
    cout << "After swap: x = " << x <<
    ", y = " << y << endl;
    return 0;
}

```

14. Difference between left shift (<<) and Right shift (>>) Operators

Left shift (<<) shifts the bits to the left. The vacated positions on the right are filled with 0s.

Right shift (>>) shifts the bits to the right. The vacated positions on the left are filled with 0s or the sign bits.

15. Cpp(Conditional Operator to Determine Larger of Two floating-point Numbers)

```

#include <iostream>
using namespace std;
int main() {
    double f1, f2;
    cout << "Enter two floating-point numbers: ";
    cin >> f1 >> f2;
    // The conditional operator
    determines the larger number
    double larger = (f1 > f2) ? f1 : f2;
}

```

```

    cout << "The larger floating-point
number is: " << larger << endl;
    return 0;
}

```

16. Difference between Bitwise Not and Logical Not Operators

Operator

Bitwise Not

Logical Not

Name

One's Complement

Negation

Symbol

~

!

Purpose

Inverts all the bits of the operand.

Inverts a boolean condition.

Example

~0101(binary) becomes 1010

!true is false

17.Cpp(Bitwise AND)

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {
    int number;
    cout << "Enter a positive integer: ";
    cin >> number;
    // check: ( number > 0) AND ((number & ( number - 1)) == 0)
    If ( number > 0 && (number & ( number - 1)) == 0) {
        cout << number << " is a power of 2. " << endl;
    } else {
        cout << number << " is Not a power of 2. " << endl;
    }
    return 0;
}

```

18. Cpp(Binary to Decimal Converter

```
#include <iostream >
```

```
#include <string>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main() {
```

```
    String binary;
```

```
    cout << "Enter a binary number
```



```
( string of 0s and 1s) : ";
    cin >> binary;
```

```
int decimal = 0;
int power = 0;
// Iterate through the binary string from right to left (LSB to MSB)
for ( int i = binary. Length() - 1; i
    >= 0; i--) {
    if ( binary[i] == '1' ) {
        decimal result
        decimal += pow(2, power);
    }
    power ++;
}
    cout << "Decimal equivalent: " <<
    decimal << endl;
    return 0;
}
```

19. Cpp(Binary to Decimal, Hexadecimal, and Octal converter)

```
#include <iostream>
#include <string>
#include <algorithm> // For reverse
using namespace std;

int main() {
    String binary;
    Cout << "Enter a binary number
    ( string of 0s and 1s) : ";
    Cin >> binary;
    // convert binary string to a decimal interger first ( using an efficient loop)
    Int decimal = 0;
    for ( char bit : binary) {
        decimal = ( decimal * 2) + (bit - '0');
        cout << "\n- - conversion Results
        — " << Decimal: standard output
        ( default )
        cout << "Decimal: " << dec <<
        decimal << endl;
        // Hexadecimal: use the hex manipulator
        cout << "Hexadecimal: " << hex <<
        uppercase << decimal << endl;
        // octal: use the oct manipulator
        cout << "octal: " << oct << decimal << endl;
        return 0;
    }
}
```